

Module 6: E Payment Mechanism

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E–Payment Mechanism

An E–Payment Mechanism refers to the electronic method of transferring money or making payments through digital systems and communication networks such as the internet, mobile networks, and banking systems. It enables customers, businesses, banks, and financial institutions to conduct financial transactions electronically without the physical exchange of cash or paper documents.

E-payment systems form the backbone of modern e-commerce because online buying and selling cannot function effectively without secure and efficient payment methods. The increasing growth of internet technology, online shopping, digital banking, and mobile commerce has significantly increased the importance of electronic payment systems.

Traditional payment methods such as cash and paper cheques involve physical handling, higher processing time, and geographical limitations. In contrast, electronic payment systems provide speed, convenience, automation, security, and global accessibility.

Objectives of E–Payment Systems

- 1) To provide fast and secure payment facilities
- 2) To reduce dependence on physical cash
- 3) To support online business and e-commerce activities
- 4) To simplify financial transactions
- 5) To reduce transaction processing cost
- 6) To improve accuracy and transparency in payments
- 7) To provide global accessibility for financial operations

Features of E–Payment Mechanisms

- 1) **Electronic Transfer of Funds:** Money is transferred electronically between customer and merchant accounts without physical cash movement.
- 2) **Speed and Efficiency:** Transactions are completed within seconds or minutes, improving business efficiency.
- 3) **Global Accessibility:** Payments can be made from any location with internet access.

- 4) **Security:** Modern e-payment systems use encryption, authentication, and digital security technologies.
- 5) **Automation:** Transaction processing, billing, and record maintenance are automated.
- 6) **Convenience:** Customers can make payments anytime and anywhere using computers or smartphones.
- 7) **Record Maintenance:** Electronic systems automatically maintain digital transaction records for future reference.

- 4) **Improved Customer Convenience:** Customers can shop and pay from their homes.
- 5) **Better Security and Monitoring:** Digital transactions can be tracked and monitored easily.
- 6) **Reduced Risk of Physical Theft:** Electronic funds reduce the need to carry physical cash.
- 7) **Enhanced Financial Management:** Digital records help businesses maintain accurate accounting and financial analysis.

Advantages of E–Payment Mechanisms

- 1) **Faster Transactions:** Electronic systems process payments quickly compared to traditional banking methods.
- 2) **Reduced Operational Cost:** Businesses save money on paperwork, cash handling, and manual processing.
- 3) **Increased Business Opportunities:** Online payment facilities help businesses operate globally.

Limitations of E–Payment Mechanisms

- 1) Dependence on internet and electricity
- 2) Risk of cybercrime and hacking
- 3) Technical failures and server issues
- 4) Digital illiteracy among some users
- 5) Privacy and data security concerns
- 6) Possibility of online fraud and identity theft

Types of E–Payment Mechanisms

- | | | |
|---------------------------------|-----------------------------|-----------------------------|
| 1) Payment through Card Systems | 3) Electronic Cash (E–Cash) | 6) UPI and Mobile Payments |
| 2) Electronic Cheque (E–Cheque) | 4) Internet Banking | 7) Digital Payment Gateways |
| | 5) Mobile Wallets | |

1) Payment through Card System

A card payment system is an electronic payment method where customers use plastic cards issued by banks or financial institutions to make online or offline purchases.

These cards contain customer account information and allow electronic authorization and transfer of funds during transactions.

Card systems are among the most widely used e-payment methods in modern commerce.

Types of Card Systems

- 1) **Credit Card:** A credit card allows customers to purchase goods and services using borrowed money provided by the issuing bank or financial institution. The customer repays the borrowed amount later according to the billing cycle.

Working Process of Credit Card Transactions

1. Customer selects goods or services online

2. Card details are entered on the payment page
3. Payment gateway securely transfers information
4. Issuing bank verifies card details and available credit limit
5. Transaction authorization is provided
6. Merchant receives payment confirmation
7. Customer later pays the bank bill

- Card Network (Visa, MasterCard, RuPay)

Advantages of Credit Cards

- Convenient online and international transactions
- Allows purchase without immediate cash payment
- Provides reward points and cashback benefits
- Useful during emergencies

Components Involved

- Cardholder
- Merchant
- Payment Gateway
- Acquiring Bank
- Issuing Bank

Disadvantages of Credit Cards

- High interest on delayed payments
- Risk of overspending
- Possibility of card fraud and theft
- Annual maintenance charges

2) Debit Card: A debit card allows customers to make payments directly from their bank account. Unlike credit cards, no borrowing facility exists.

Working Process

1. Customer enters debit card details
2. Bank checks account balance
3. Amount is deducted instantly
4. Merchant receives payment confirmation

- Better spending control
- Lower risk of debt accumulation
- Easy ATM and online usage

Advantages

- Immediate payment deduction

Limitations

- Requires sufficient account balance
- Limited credit facility
- Online fraud risks still exist

3) Smart Card: A smart card is an advanced electronic card containing an embedded microprocessor chip capable of storing and processing information securely.

Features of Smart Cards

- Data storage capability
- Encryption support
- High security
- Multi-purpose applications

- Transportation systems
- Access control systems

Advantages

- Better security than magnetic stripe cards
- Reduced fraud possibilities
- Faster authentication

Applications

- Banking systems
- Identity verification

Security in Card Payment Systems

Card systems use various security technologies:

- CVV number verification
- OTP (One Time Password)
- SSL/TLS encryption
- EMV chip technology
- Tokenization systems

2) E–Cheque (Electronic Cheque)

An E–Cheque is the electronic version of a traditional paper cheque used for transferring funds digitally between bank accounts. It performs similar functions as a paper cheque but uses electronic communication and digital signatures for authentication.

Components of E–Cheque

- Payer information
- Payee information
- Bank account details
- Payment amount
- Digital signature
- Transaction date

Working Process of E–Cheque

1. Customer creates an electronic cheque
2. Digital signature is attached
3. E-cheque is transmitted electronically
4. Merchant sends it to the bank
5. Bank verifies digital signature and account details
6. Funds are transferred electronically
7. Confirmation message is generated

Features of E–Cheque

- Paperless transaction system
- Secure digital authentication
- Faster cheque processing
- Electronic verification mechanism

Advantages of E–Cheque

- 1) **Reduced Paperwork:** No physical cheque handling is required.
- 2) **Faster Processing:** Electronic transfer reduces processing time.
- 3) **Lower Administrative Cost:** Banks and businesses save operational expenses.
- 4) **Better Security:** Digital signatures reduce cheque forgery.
- 5) **Easy Record Maintenance:** Electronic storage improves tracking and auditing.

Disadvantages of E–Cheque

- Requires internet and banking infrastructure
- Technical knowledge is necessary
- Vulnerable to cyber-attacks if poorly secured
- Limited adoption in some areas

Applications of E–Cheque

- Online bill payment
- Business-to-business transactions
- Salary and vendor payments
- Banking operations

3) E–Cash (Electronic Cash)

E–Cash refers to digital money stored electronically and used for online transactions. It acts like physical cash but exists only in digital form. Electronic cash systems allow users to transfer digital monetary value through electronic networks.

Characteristics of E–Cash

- Digital monetary value
- Instant payment transfer
- Electronic storage
- Internet-based transactions
- High transaction speed

Working Process of E–Cash

1. User deposits money into digital wallet
2. Electronic tokens or digital value are generated

3. User sends e-cash to merchant during purchase
4. Merchant verifies payment authenticity
5. Transaction is completed electronically

Types of E–Cash

1) Identified E–Cash: Identity of the user is known and recorded during transactions.

Advantages

- Better security
- Easier fraud detection

2) Anonymous E–Cash: User identity is hidden during transactions.

Advantages

- Improved privacy

Disadvantages

- May increase illegal transaction risks

Advantages of E–Cash

- 1) **Fast Transactions:** Instant online payment processing.
- 2) **Convenient Usage:** Suitable for online shopping and digital services.
- 3) **Reduced Handling Cost:** No physical currency handling required.
- 4) **Suitable for Micropayments:** Useful for small online payments.
- 5) **Global Accessibility:** Can be used internationally through internet systems.

Limitations of E–Cash

- Risk of hacking and cyber theft
- Dependence on internet connectivity
- Technical infrastructure requirements
- Limited acceptance in some locations

Applications of E–Cash

- Online shopping
- Mobile wallets
- Digital gaming purchases
- Subscription services
- Utility bill payments

4) Internet Banking

Internet Banking, also known as Online Banking or E-Banking, refers to banking services provided through the internet. It allows customers to access and manage their bank accounts electronically using computers, smartphones, or other internet-enabled devices. Customers can perform banking activities without physically visiting the bank branch.

Objectives of Internet Banking

- To provide banking services anytime and anywhere
- To reduce dependency on physical bank branches
- To improve speed and efficiency of banking operations
- To support electronic commerce and digital payments

- To provide convenient financial management services

Features of Internet Banking

1. **Online Account Access:** Customers can view account balance, account statements, and transaction history.

2. **Fund Transfer Facility:** Money can be transferred between accounts electronically.
3. **Bill Payment:** Utility bills, mobile recharges, taxes, and subscriptions can be paid online.
4. **Online Shopping Payments:** Internet banking supports e-commerce transactions.
5. **Banking Services:** Customers can request cheque books, fixed deposits, loans, and account updates.
6. **24×7 Availability:** Services are available continuously through the internet.

Working Process of Internet Banking

1. Customer registers for internet banking services
2. Bank provides login credentials and authentication tools
3. Customer logs into the secure banking portal
4. Banking requests are processed electronically
5. Transaction confirmation is generated

Services Provided through Internet Banking

- Fund transfer (NEFT, RTGS, IMPS)
- Account balance enquiry
- Online bill payment
- Loan applications
- Investment services
- Tax payment
- Mobile recharge

5) Mobile Wallets

A Mobile Wallet, also called a Digital Wallet or E-Wallet, is an electronic application that stores payment information digitally and allows users to make electronic transactions through smartphones or mobile devices. It stores money electronically and enables cashless payments.

Examples of Mobile Wallets

- Paytm
- PhonePe
- Google Pay

- Statement downloads

Advantages of Internet Banking

- 1) **Convenience:** Customers can perform banking activities from home or office.
- 2) **Time Saving:** Eliminates waiting time at bank branches.
- 3) **Faster Transactions:** Electronic processing speeds up financial operations.
- 4) **Reduced Operational Cost:** Banks reduce paperwork and staffing requirements.
- 5) **Better Financial Management:** Digital records help track transactions easily.

Disadvantages of Internet Banking

- Dependence on internet connectivity
- Cybersecurity risks and hacking
- Technical failures and server downtime
- Requires digital literacy
- Risk of phishing attacks

Security Measures in Internet Banking

- SSL/TLS encryption
- OTP verification
- Two-factor authentication
- Biometric authentication
- Secure login credentials
- Firewalls and antivirus systems

Applications of Internet Banking

- Personal banking
- Business transactions
- Online shopping
- Government tax payments
- International fund transfers

- Amazon Pay
- Mobikwik

Features of Mobile Wallets

- 1) **Digital Storage of Money:** Money can be stored electronically within the wallet.
- 2) **Quick Payments:** Payments can be made instantly using smartphones.
- 3) **QR Code Scanning:** Users can scan QR codes for payment.
- 4) **Recharge and Bill Payment:** Supports mobile recharge, utility bills, and ticket booking.
- 5) **Integration with Bank Accounts:** Wallets can be linked with bank accounts and cards.

Working Process of Mobile Wallets

1. User downloads wallet application
2. Wallet account is created
3. Money is added through bank transfer or card
4. User makes payments through wallet balance or linked account
5. Merchant receives electronic confirmation

Types of Mobile Wallets

- 1) **Closed Wallet:** Used only for transactions with a specific company. **Example:** Amazon Pay balance for Amazon purchases.
- 2) **Semi-Closed Wallet:** Allows transactions at approved merchants. **Examples:** Paytm Wallet, Mobikwik.

- 3) **Open Wallet:** Allows purchase, fund transfer, and cash withdrawal. **Examples:** Bank-issued wallets.

Advantages of Mobile Wallets

- 1) **Convenience:** Easy and fast mobile-based transactions.
- 2) **Cashless Payments:** Reduces dependence on physical cash.
- 3) **Instant Transactions:** Payments occur in real-time.
- 4) **Rewards and Cashback:** Many wallets offer promotional benefits.
- 5) **Secure Transactions:** Authentication systems improve security.

Disadvantages of Mobile Wallets

- Dependence on smartphones and internet
- Battery and device limitations
- Security risks if device is stolen
- Transaction limits may exist

Security Measures in Mobile Wallets

- PIN and password protection
- Fingerprint and biometric authentication
- OTP verification
- Data encryption
- Tokenization technology

Applications of Mobile Wallets

- Retail payments
- Online shopping
- Utility bill payments
- Transportation ticket booking
- Food delivery payments

6) UPI and Mobile Payments

UPI (Unified Payments Interface) is a real-time digital payment system developed by the National Payments Corporation of India (NPCI) that allows instant money transfer between bank accounts using mobile devices. It integrates multiple bank accounts into a single mobile application.

Features of UPI

- 1) **Real-Time Fund Transfer:** Money transfer occurs instantly.
- 2) **24×7 Availability:** Transactions are available anytime.
- 3) **Bank-to-Bank Transfer:** Direct transfer between bank accounts.
- 4) **Single Mobile Application:** Multiple bank accounts can be managed in one app.
- 5) **Virtual Payment Address (VPA):** Transactions can occur without sharing account details.

Working Process of UPI

1. User installs UPI-enabled application
2. Bank account is linked to mobile number
3. UPI PIN is created
4. User enters recipient VPA or scans QR code
5. Payment authorization is completed
6. Funds transfer instantly

Examples of UPI Applications

- Google Pay
- PhonePe
- BHIM
- Paytm
- Amazon Pay

Advantages of UPI

- 1) **Instant Money Transfer:** Transactions are processed immediately.
- 2) **Easy to Use:** Simple mobile interface.
- 3) **No Need for Account Details:** VPA simplifies transactions.
- 4) **Low Transaction Cost:** Most UPI payments are free.
- 5) **Supports Financial Inclusion:** Encourages digital banking adoption.

Limitations of UPI

- Requires internet connectivity
- Server overload may delay transactions
- Fraud and phishing risks
- Daily transaction limits

Mobile Payments

Mobile payment refers to financial transactions conducted using mobile devices such as smartphones or tablets. It includes payments through UPI, wallets, NFC technology, QR codes, and mobile banking applications.

Types of Mobile Payments

- 1) **SMS-Based Payments:** Transactions are conducted through SMS services.
- 2) **NFC Payments:** Near Field Communication enables contactless payments.
- 3) **QR Code Payments:** Scanning QR codes for payment transfer.
- 4) **App-Based Payments:** Payments through mobile applications.

Advantages of Mobile Payments

- High convenience
- Contactless transactions
- Faster payment processing
- Reduced cash handling
- Easy transaction tracking

Security Measures in UPI and Mobile Payments

- UPI PIN authentication
- Device binding security
- End-to-end encryption
- OTP verification
- Fraud detection systems

7) Digital Payment Gateways

A Digital Payment Gateway is a technology platform that securely processes online payments between customers, merchants, and financial institutions.

It acts as an intermediary that transfers payment information securely during e-commerce transactions.

Functions of Payment Gateways

- Authorization of payments
- Encryption of transaction data
- Communication between banks and merchants
- Fraud detection and prevention
- Transaction confirmation

Working Process of Payment Gateway

1. Customer selects product online
2. Payment details are entered
3. Payment gateway encrypts data
4. Information is sent to acquiring bank
5. Issuing bank verifies transaction
6. Authorization response is generated
7. Merchant receives payment confirmation

Components of Payment Gateway System

- Customer
- Merchant website
- Payment gateway
- Acquiring bank
- Issuing bank
- Card network

Features of Digital Payment Gateways

- 1) **Secure Transactions:** Uses encryption and authentication technologies.
- 2) **Multiple Payment Options:** Supports cards, wallets, UPI, and net banking.
- 3) **Real-Time Processing:** Transactions occur instantly.

- 4) **Fraud Detection:** Advanced monitoring systems identify suspicious activities.
- 5) **International Payment Support:** Supports global e-commerce transactions.

Advantages of Payment Gateways

- 1) **Secure Online Payments:** Protects sensitive customer information.
- 2) **Faster Transaction Processing:** Improves customer experience.
- 3) **Supports E-Commerce Growth:** Enables businesses to accept digital payments.
- 4) **Automation:** Automatic payment verification and record generation.
- 5) **Multiple Currency Support:** Useful for international business.

Disadvantages of Payment Gateways

- Service charges and transaction fees
- Technical failures may interrupt transactions
- Cybersecurity risks
- Dependence on internet systems

Security Technologies Used in Payment Gateways

- SSL/TLS encryption
- Tokenization
- PCI DSS compliance
- Fraud monitoring systems
- Two-factor authentication

Examples of Payment Gateways

- Razorpay
- PayU
- PayPal
- Stripe
- CCAvenue

Importance of Digital Payment Systems in E-Commerce

- Enables online business transactions
- Improves customer convenience
- Supports cashless economy
- Facilitates global trade
- Increases transaction efficiency

E-Payment Threats and Protections

E-Payment Threats

Electronic payment systems face several security threats because financial transactions occur through digital networks.

- 1) **Hacking:** Unauthorized access to banking systems, payment gateways, or customer accounts. **Effects**
 - Financial loss
 - Data theft
 - System damage
- 2) **Phishing:** Fraudulent emails, websites, or messages are used to steal sensitive information such as passwords and card details. **Example:** Fake banking websites asking for login credentials.
- 3) **Identity Theft:** Cybercriminals steal personal and banking information to perform unauthorized transactions.
- 4) **Malware and Viruses:** Malicious software infects devices and captures confidential information. **Examples**
 - Keyloggers
 - Trojan viruses
 - Spyware
- 6) **Card Fraud:** Illegal use of stolen card details for unauthorized purchases.
- 7) **Data Breach:** Sensitive customer and financial information is leaked due to weak security systems.
- 8) **Man-in-the-Middle Attack:** An attacker secretly intercepts communication between customer and payment server.
- 9) **Denial of Service (DoS) Attack:** Attackers overload payment servers, causing service disruption.

Protection Mechanisms in E-Payment Systems

- 1) **Encryption:** Encryption converts sensitive information into coded form to prevent unauthorized access. **Common Technologies**
 - SSL (Secure Socket Layer)
 - TLS (Transport Layer Security)
- 2) **Digital Signature:** Digital signatures verify sender identity and ensure transaction authenticity. **Benefits**
 - Authentication
 - Non-repudiation
 - Data integrity
- 3) **Firewalls:** Firewalls monitor and control incoming and outgoing network traffic to prevent unauthorized access.
- 4) **Two-Factor Authentication (2FA):** Requires two verification methods such as password and OTP. **Advantages**
 - Increased account security
 - Reduced unauthorized access

5) Secure Payment Gateway: A payment gateway securely processes payment information between customer and merchant. **Functions**

- Transaction authorization
- Data encryption
- Fraud prevention

6) Tokenization: Sensitive card details are replaced with random secure tokens.

Benefits

- Protects real card information
- Reduces fraud risk

7) Antivirus and Anti-Malware Systems: Protect devices and servers from malicious software attacks.

8) Biometric Authentication: Uses fingerprint, facial recognition, or iris scanning for user verification

Importance of Security in E–Payment Systems

Security is essential because electronic payment systems handle sensitive financial information.

Importance

- Protects customer data
- Prevents financial fraud
- Increases customer confidence
- Supports safe online commerce
- Ensures business continuity

Future Trends in E–Payment Systems

- Artificial Intelligence based fraud detection
- Blockchain-based payment systems
- Biometric payment authentication
- Contactless payments
- Cryptocurrency integration
- Mobile and wearable payments